

ECEN 4/5632 Theory and Application of Digital Filters

Lecture 4, 9-03-26

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* With generous help from Gemini Pro and ChatGPT

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Deutsch

Wagon-wheel effect

from Michael's Visual Phenomena & Optical Illusions



What to observe

When this page has loaded, the neighboring demo is in "auto run" mode: the wheel rotation smoothly varies between standing still and a maximum speed of 100 rotations per minute (100 rpm, indicated at the top right). The wheel rotation, however, looks quite different: it seems to slow down to standstill when the maximum speed of 100 rpm is indicated! It helps to know that the wheel is rendered with 20 frames per second.



©MBach

150 rpm

Stop

Reset

☐ Autorun

☐ Mark 1 spoke

Built with...

©2007–26 MB...

Doesn't work?

$$150 \text{ rpm} = 2.5 \text{ rps}$$

12 spokes

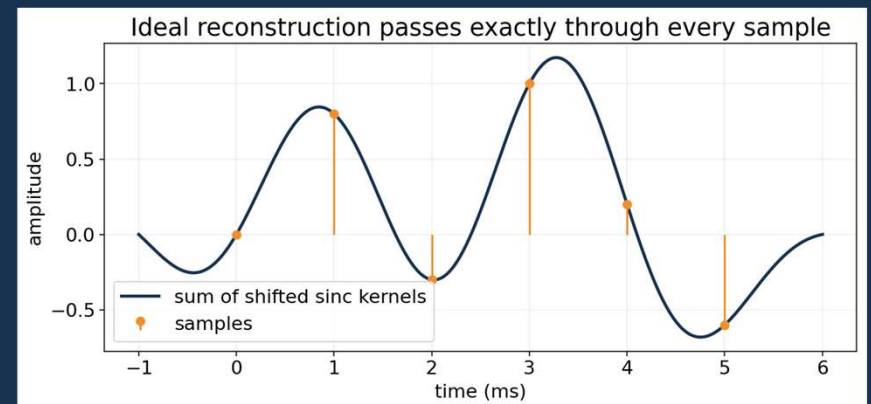
$$12 \times 2.5 = 30$$

⇒ display
refresh rate
is 30 Hz
or perhaps
60 Hz

application: Measure
speed of turbine

Reconstruction, D/A, and Quantization Noise

Ideal interpolation gives the theory; ZOH and finite bit depth explain what hardware actually does.



75 minutes • ideal D/C • sinc interpolation • zero-order hold • quantization • SQNR • PyTorch dtype bridge

Recap: after Lecture 3, we trust the sample times

Now we ask how to reconstruct between samples and represent amplitudes.

Sampling error

Where Week 2 lives in the audio → ML pipeline

Quantization error



Aliasing moves energy to the wrong frequency.
It is fundamentally non-unique.

Sampling
CT → DT

Quantization maps a continuous amplitude to one of finitely many levels.

quantization
continuous → discrete amplitude

Learning outcomes

By the end of Lecture 4, students should be able to move between ideal theory and real hardware.

- Derive the ideal bandlimited interpolation formula from the reconstruction low-pass filter.
- Explain zero-order hold behavior in time and its sinc-shaped magnitude droop.
- Model a uniform B-bit quantizer and compute Δ and quantization-error variance.
- Derive and use SQNR $\approx 6.02B + 1.76$ dB for a full-scale sine.
- Explain the difference between 16-bit PCM storage and float32 computation in PyTorch.

Ideal reconstruction in the frequency domain

If spectral replicas do not overlap, simply isolate the baseband copy.

$$X_s(j\Omega) = \frac{1}{T} \sum_k X_c(j(\Omega - k\Omega_s))$$

$$H_r(j\Omega) = \begin{cases} T, & |\Omega| < \Omega_c \\ 0, & \text{otherwise} \end{cases}$$

The factor T cancels the 1/T introduced by ideal sampling.

Choose $\Omega_M < \Omega_c < \Omega_s - \Omega_M$ so only one replica passes.

$$x[n] = x_s(nT) = x_c(nT) \sum_n \delta(t - nT)$$

from $x[n]$ back to $x_r(t) \stackrel{\text{hopefully}}{=} x_c(t)$

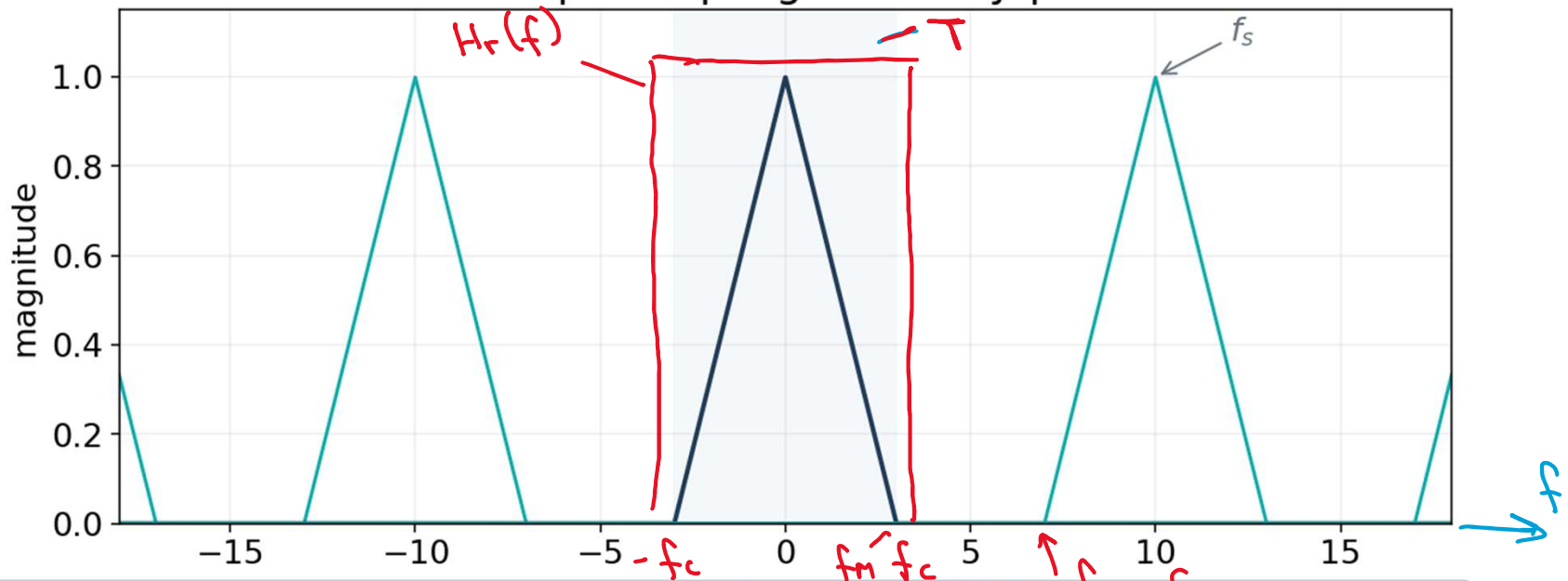
$$f_s = \frac{1}{T} \text{ sampling rate}$$

What the sampled spectrum looks like

Every copy contains the same analog information—until the copies overlap.

$H_r(j\Omega) = \begin{cases} T & \Omega < \Omega_c \\ 0 & \text{otherwise} \end{cases}$
 brickwall filter

No overlap: sampling above Nyquist rate



Replica spacing = f_s . Baseband can be isolated only when neighboring replicas do not overlap.

$$f_m < f_c < f_s - f_m$$

$f_s/2$ satisfies conditions

From ideal low-pass filtering to sinc interpolation

The inverse transform of a brick-wall low-pass filter is a sinc.

$$\text{sinc}(u) \triangleq \frac{\sin(\pi u)}{\pi u}$$

$$\underline{h_r(t) = \text{sinc}\left(\frac{t}{T}\right)} \quad \text{for the ideal Nyquist-band reconstruction case}$$

$$H_r(j\Omega) = \begin{cases} T & |\Omega| < \frac{\Omega_s}{2} \\ 0 & \text{otherwise} \end{cases}$$

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n] \text{sinc}\left(\frac{t - nT}{T}\right)$$

$$\Omega_s = 2\pi f_s = \frac{2\pi}{T}$$

$$\Omega_s/2 = \frac{\pi}{T}$$

Here $\text{sinc}(u) = \sin(\pi u)/(\pi u)$. Every sample contributes one shifted, scaled interpolation kernel.

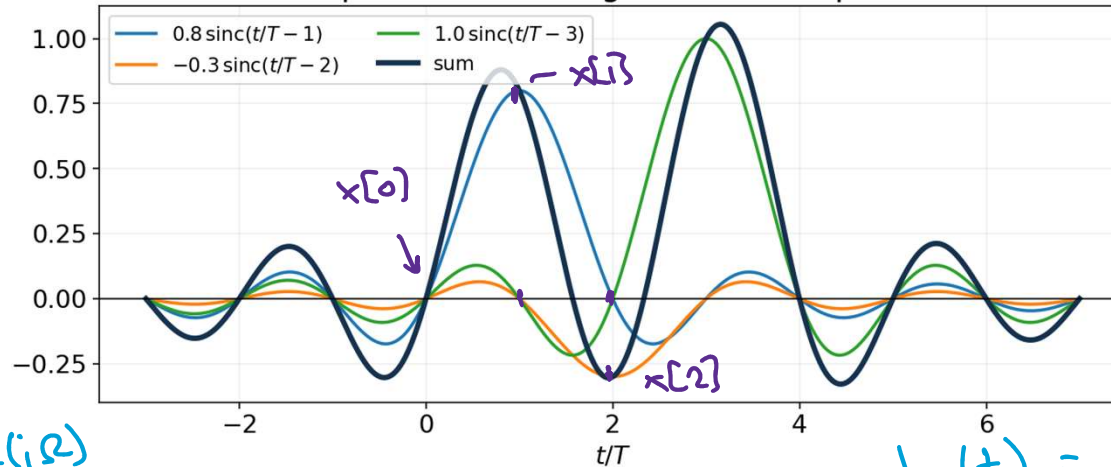
$$\begin{aligned} h_r(t) &= \frac{1}{2\pi} \int_{-\Omega_s/2}^{\Omega_s/2} T e^{j\Omega t} d\Omega = \frac{T}{2\pi} \left. \frac{e^{j\Omega t}}{jt} \right|_{-\Omega_s/2}^{\Omega_s/2} = \frac{\Omega_s T}{2\pi} \frac{e^{j\Omega_s t/2} - e^{-j\Omega_s t/2}}{\Omega_s jt/2} \\ &= \frac{\Omega_s T}{\pi} \frac{\sin \Omega_s t/2}{\Omega_s t/2} = \frac{2\pi}{T} \frac{T}{\pi} \frac{\sin(\pi t/T)}{\pi t/T} = \frac{\sin \pi t/T}{\pi t/T} = \text{sinc}\left(\frac{t}{T}\right) \end{aligned}$$

too much

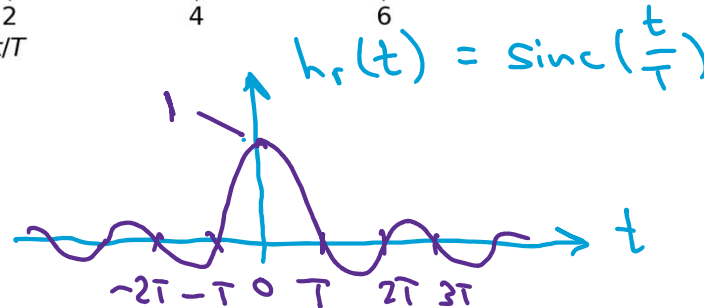
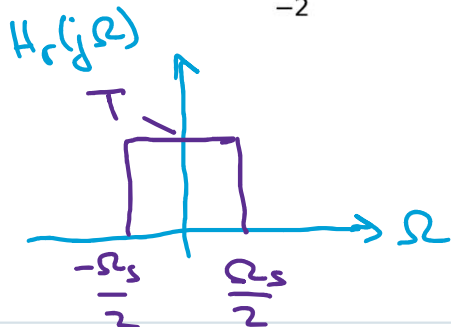
Why the sinc basis passes through every sample

At integer offsets, normalized sinc is zero—except at the center.

Interpolation is a weighted basis expansion



At $t=mT$, only the $n=m$ kernel survives $\rightarrow x_r(mT)=x[m]$.



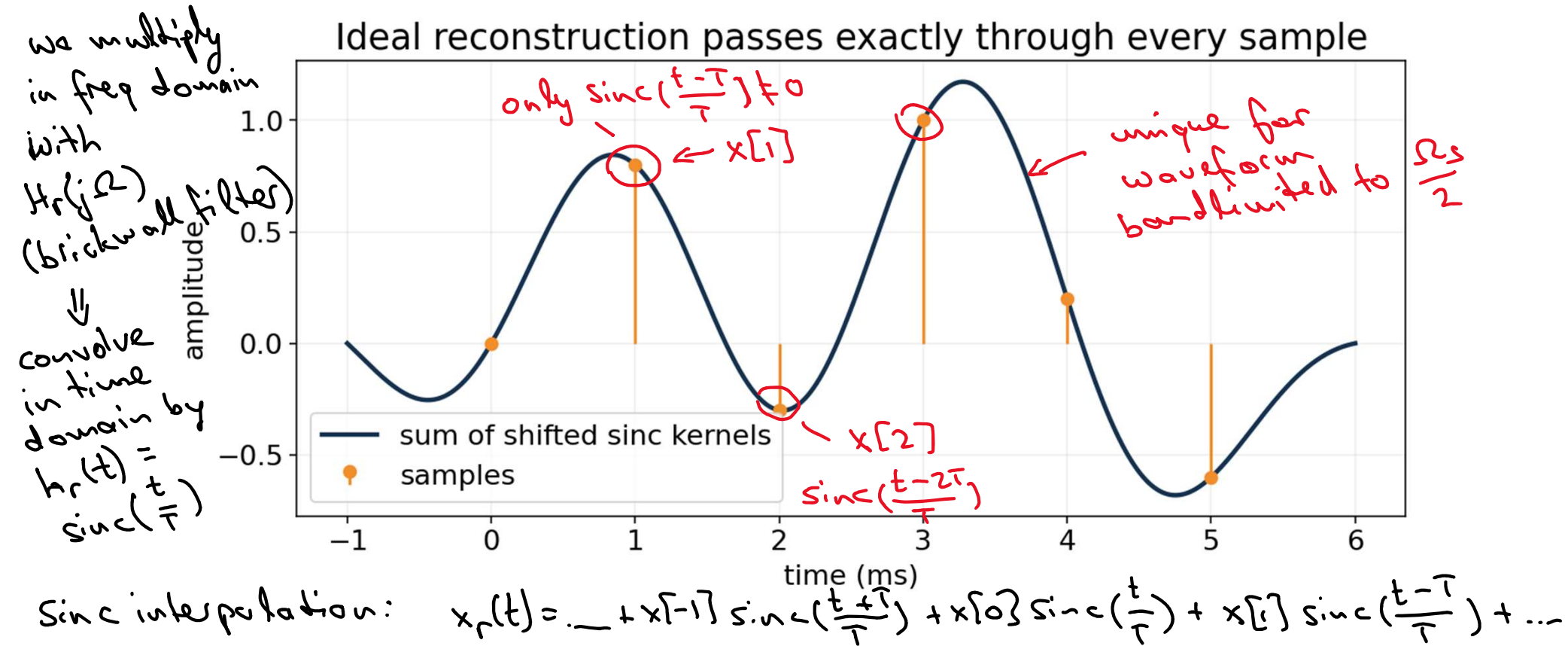
$$\text{sinc}\left(\frac{t}{T}\right) = 1 \text{ if } t=0$$

$$= 0 \text{ if } t=kT$$

$$\frac{\sin(\pi t/T)}{\pi t/T}$$

A finite illustration of ideal interpolation

The true ideal sum is infinite; a finite example still shows the geometry.



Why a real DAC does not implement ideal sinc directly

The ideal impulse response is infinite in duration and noncausal when centered at $t=0$.

Ideal mathematical interpolator

- Infinite support in time.
- Symmetric around the interpolation instant.
- Brick-wall frequency response.
- Perfect only under ideal bandlimiting assumptions.

Practical hardware strategy

- Generate a simple analog waveform from each code.
- Zero-order hold is extremely natural in a DAC.
- Follow with an analog reconstruction / anti-imaging filter.
- Oversampling reduces the burden on analog filtering.

Why do we not use $\text{sinc}(\frac{t}{T})$ in practice?

why not truncate

Gibb's phenomenon

height of overshoot waves decreases if sinc is truncated

overshoot

Ideal theory tells us the target; implementation approximates it.

Illustration of the Gibbs Phenomenon

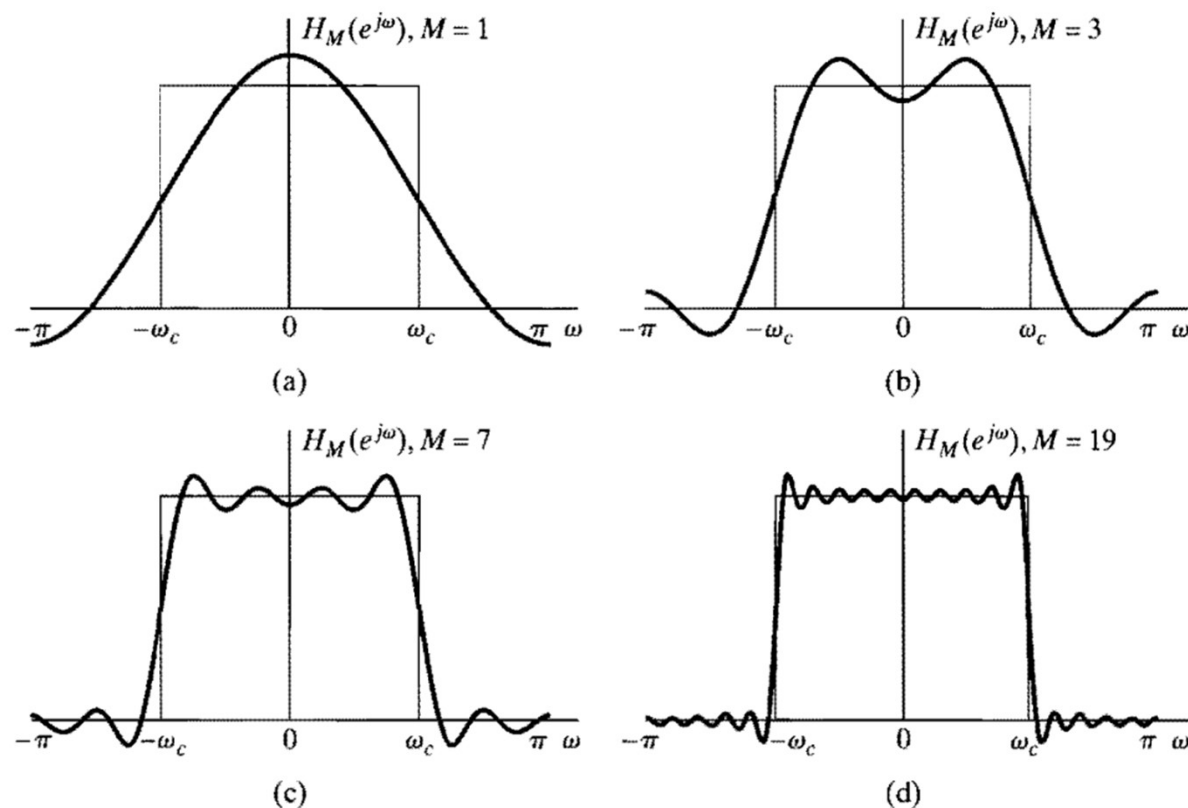


Figure 2.21 Convergence of the Fourier transform. The oscillatory behavior at $\omega = \omega_c$ is often called the Gibbs phenomenon.

DTFT of truncated (to M) “sinc” impulse response

$$H_M(e^{j\omega}) = \sum_{n=-M}^M \frac{\sin(\omega_c n)}{\pi n} e^{-j\omega n}$$

The overshoot is approximately 9% of the total height and does not decrease as M increases. Even the infinite sum ($M = \infty$) does not converge uniformly to the discontinuous function

$$H_{lp}(e^{j\omega}) = \begin{cases} 1, & |\omega| < \omega_c, \\ 0, & \omega_c < |\omega| \leq \pi. \end{cases}$$

(Brickwall filter)

Zero-order hold (ZOH): hold each sample for one period

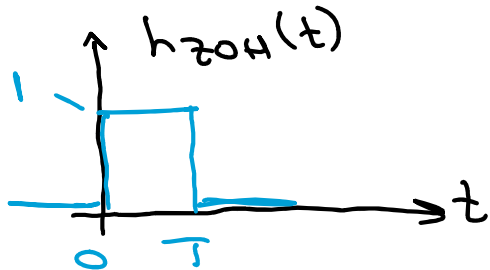
A rectangular pulse is convolved with the weighted impulse train.

$$h_{ZOH}(t) = u(t) - u(t - T)$$

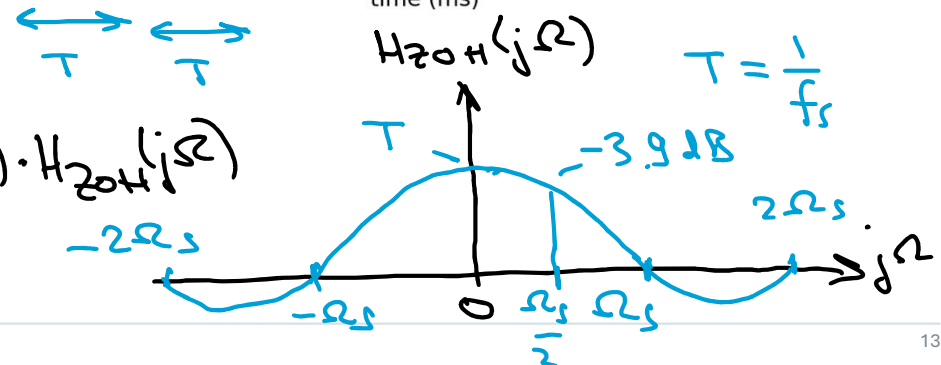
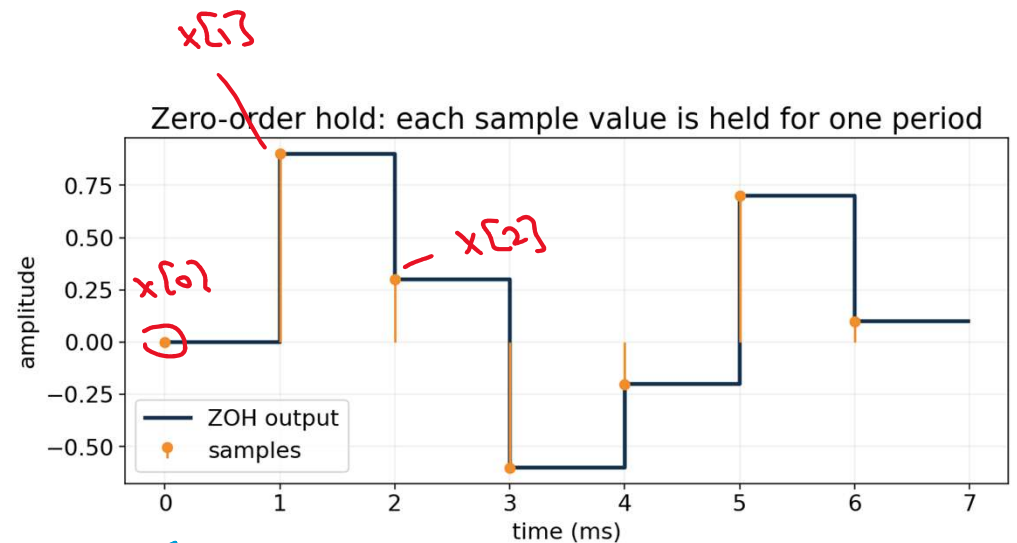
$$x_{ZOH}(t) = x_s(t) * h_{ZOH}(t)$$

conv in time

This staircase is already a continuous-time waveform—but it contains spectral images and amplitude droop.



$$X_{ZOH}(j\Omega) = X_s(j\Omega) \cdot H_{ZOH}(j\Omega)$$



Deriving the ZOH frequency response

Transform the rectangular hold pulse.

$$\underline{H_{ZOH}(j\Omega) = \int_0^T e^{-j\Omega t} dt}$$

$$= T e^{-j\Omega T/2} \frac{\sin(\Omega T/2)}{\Omega T/2}$$

delay by $T/2$

$$\omega = \frac{\Omega}{2} = \frac{\pi}{T}$$

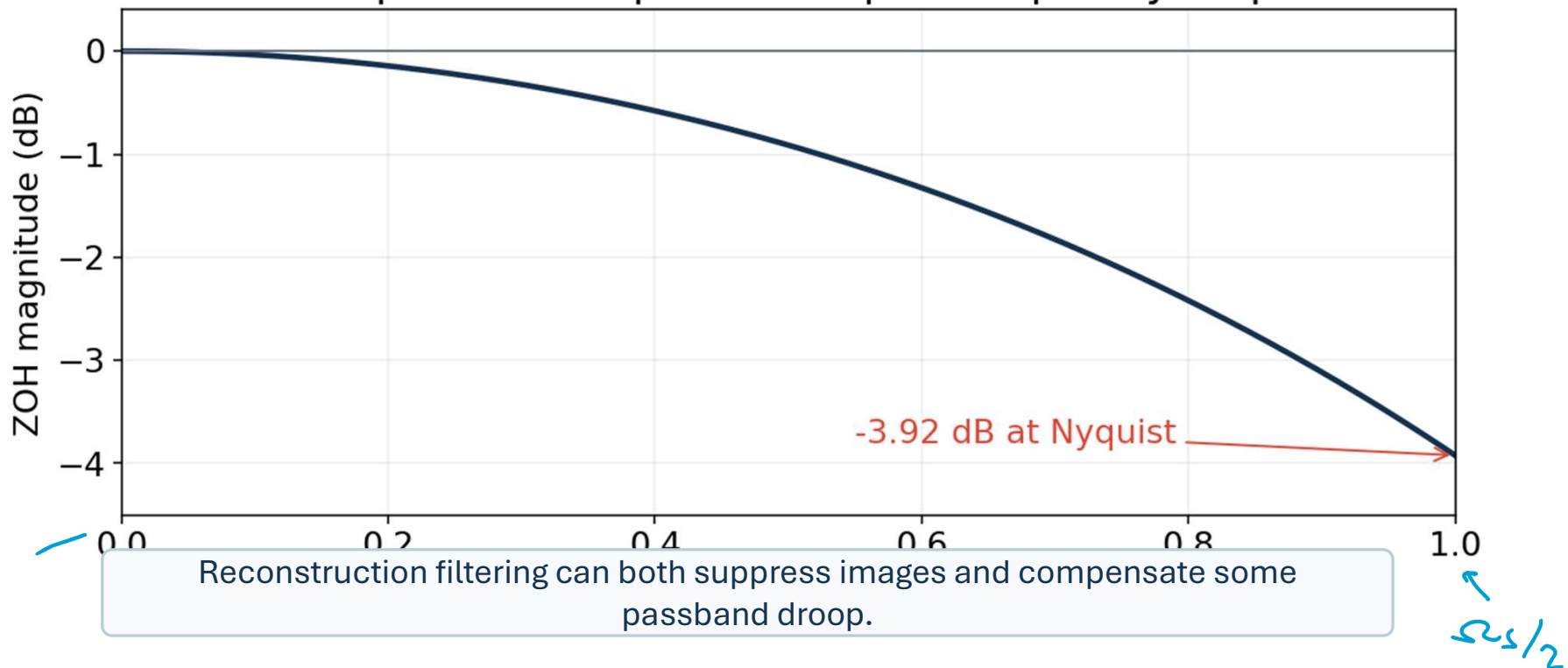
- Linear phase term: $e^{-j\Omega T/2}$ corresponds to a $T/2$ delay.
- Magnitude term: sinc-shaped envelope.
- High frequencies are attenuated more than low frequencies.

At $\Omega T = \pi$: $|H|/T = 2/\pi \approx \underline{0.637} \rightarrow \underline{-3.92 \text{ dB}}$.

The ZOH creates sinc-shaped amplitude droop

This is deterministic frequency response—not random noise.

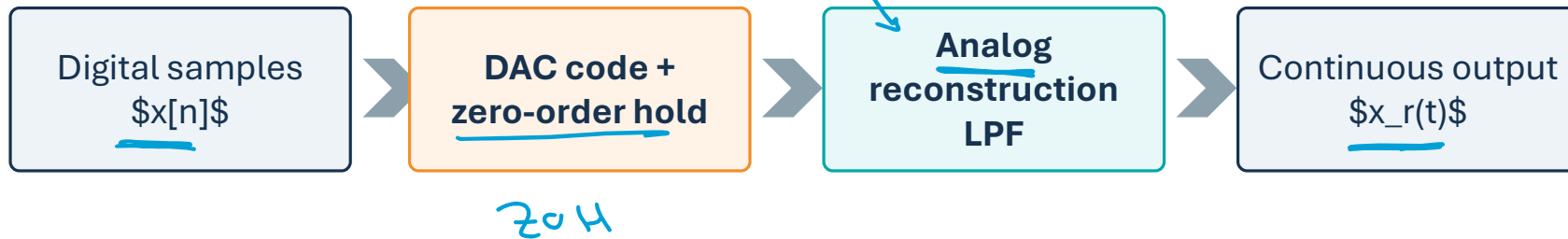
ZOH amplitude droop: sinc-shaped frequency response



A practical D/A chain

ZOH is only one stage; analog filtering completes the reconstruction.

alleviated if sampling rate is increased before D/A conversion



The analog LPF removes spectral images around multiples of f_s ; oversampling gives the filter more transition width.

Quantization: the second discretization

Sampling discretizes time. Quantization discretizes amplitude.

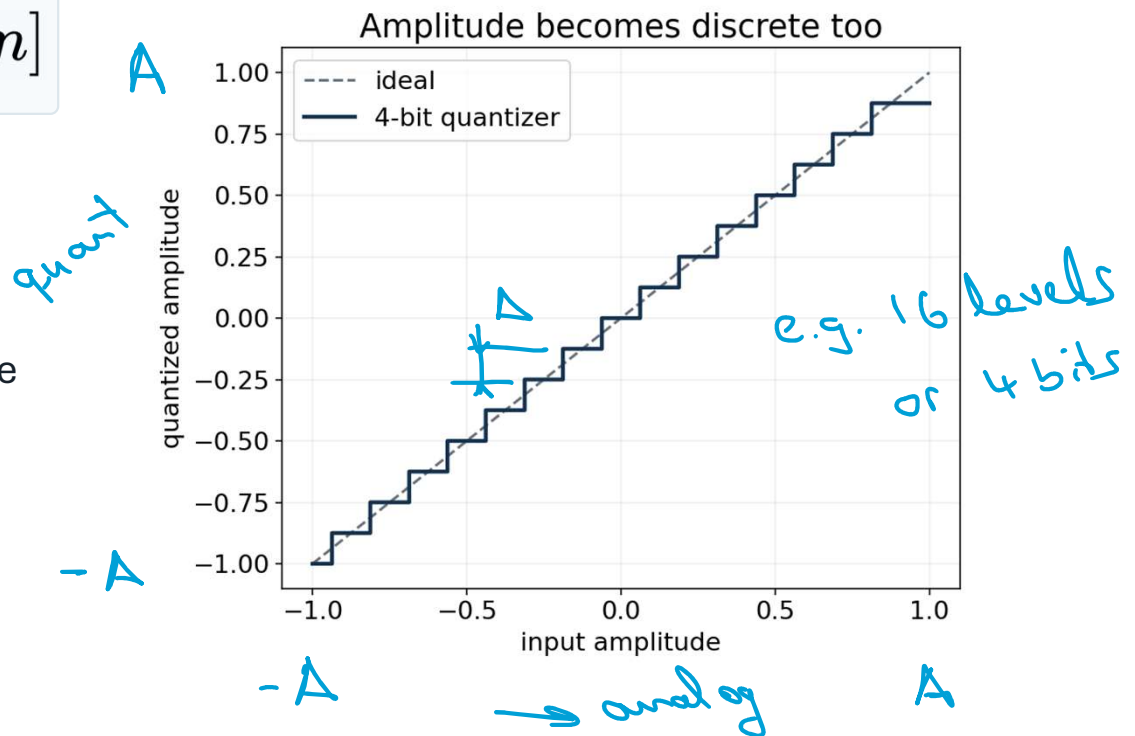
$$x_q[n] = Q\{x[n]\} = x[n] + e[n]$$

$$\Delta = \frac{2A}{2^B}$$

- B bits $\rightarrow 2^B$ nominal levels.
- Δ is the quantizer step for full-scale range roughly $[-A, A]$.
- Overload/clipping is a separate failure mode when $|x|$ exceeds the designed range.

$$SQNR = \frac{P_x}{P_e}$$

↑ error from quantization



The additive quantization-noise model

Useful—but only under assumptions.

$$e[n] = x_q[n] - x[n]$$

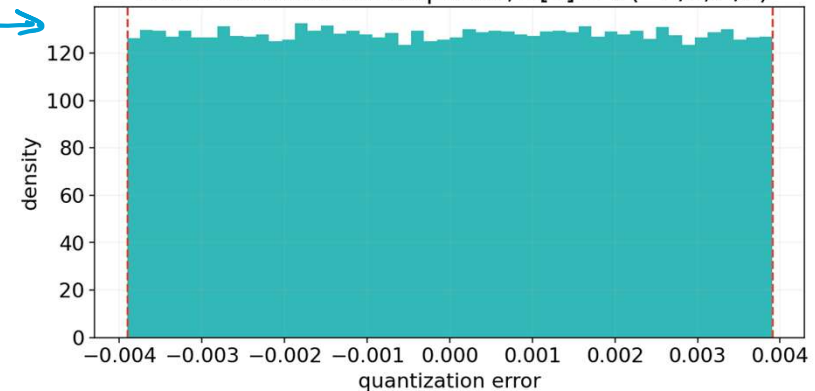
$$e[n] \approx \mathcal{U}(-\Delta/2, \Delta/2)$$

$$\mathbb{E}\{e[n]\} \approx 0, \quad \sigma_e^2 \approx \frac{\Delta^2}{12}$$

uniform
dist.
in $[-\Delta/2, +\Delta/2]$

in practice quant error e
is reasonably uniform
if $B \geq 6$ bits

Under common assumptions, $e[n] \approx \mathcal{U}(-\Delta/2, \Delta/2)$



Good approximation when the signal spans many levels and the error is sufficiently decorrelated from the signal.

Derive the quantization-noise variance

Uniform error over one quantization interval.

$$p_e(e) = \frac{1}{\Delta}, \quad -\Delta/2 \leq e < \Delta/2$$

$$\sigma_e^2 = \int_{-\Delta/2}^{\Delta/2} e^2 \frac{1}{\Delta} de = \frac{\Delta^2}{12}$$

← variance of quant error
= power of quant error

Every additional bit halves Δ → noise power falls by a factor of 4 → about 6 dB improvement.

Full-scale sine SQNR: where $6.02B + 1.76$ dB comes from

This is not an arbitrary rule of thumb; it follows from signal and error power.

$$P_x = \frac{A^2}{2}, \quad P_e = \frac{\Delta^2}{12}, \quad \Delta = \frac{2A}{2^B}$$

$$\text{SQNR} = 10 \log_{10} \frac{P_x}{P_e} = 10 \log_{10} \left(\frac{3}{2} 2^{2B} \right)$$

Signal power (pointing to P_x)
quant error power (pointing to P_e)

$$\text{SQNR} \approx 6.02B + 1.76 \text{ dB}$$

$$10 \log_{10} \left(\frac{3}{2} \right)$$

The +1.76 dB constant is specific to a full-scale sine and the ideal uniform-quantizer model.

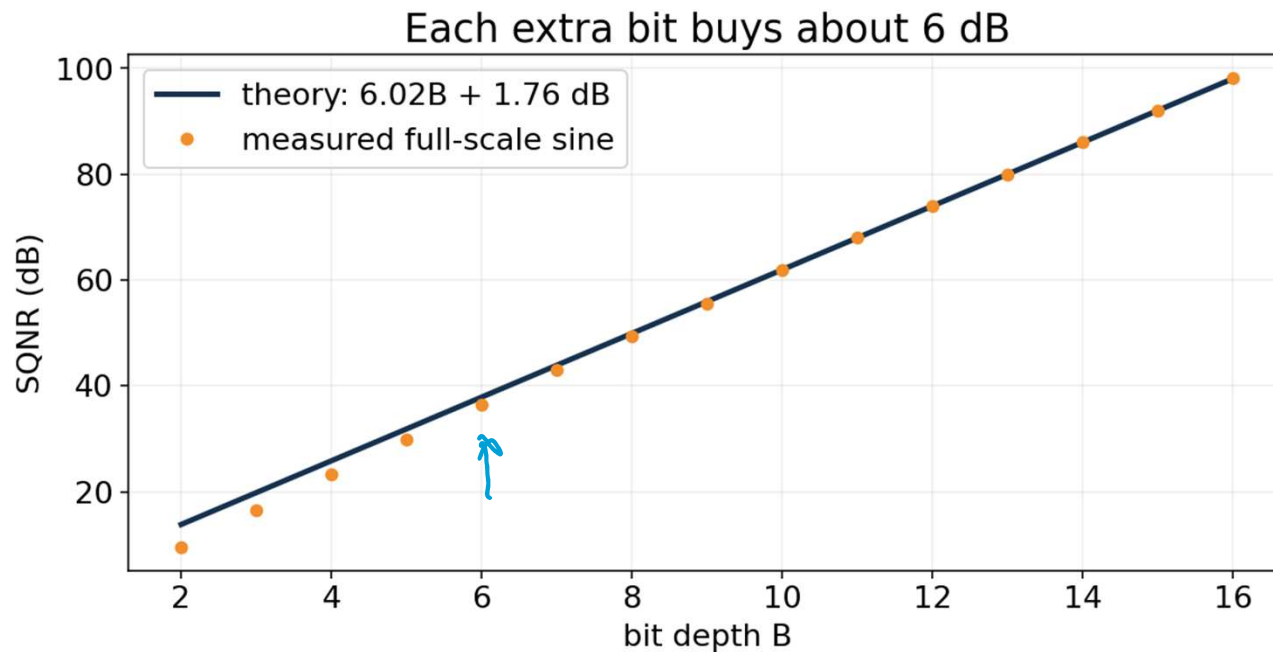
$$\Delta = \frac{2A}{2^B} \quad \text{if } B \rightarrow B+1 \Rightarrow \frac{\Delta}{2} \Rightarrow P_e^{B+1} = \frac{1}{4} \text{ of } P_e^B$$

$$10 \log_{10} \frac{1}{4} \approx -6 \text{ dB}$$

translates into +6 dB of SQNR

About 6 dB per bit—verified numerically

The simple model is remarkably accurate at moderate/high resolution.



4-bit results deviate more because the high-resolution assumptions are weak.

16-bit full-scale sine theory: 98.08 dB SQNR.

16-bit audio: “96 dB” and “98 dB” can both appear

They refer to slightly different reference calculations.

6 dB/bit rule of thumb

$$20 \log_{10}(2^{16}) \approx \underline{96.33 \text{ dB}}$$

16 × 6.02

96.3 dB is the amplitude ratio across 16 binary bits.

Full-scale sine SQNR

$$6.02(16) + \underline{1.76} \approx 98.08 \text{ dB}$$

specific to sine full amplitude

98.1 dB includes the RMS power of a full-scale sine and uniform error model.

Be precise about what quantity is being compared.

PCM storage is not the same thing as ML compute dtype

Casting to float changes representation—not the information already captured.

```
pcm = np.array(..., dtype=np.int16)
x = pcm.astype(np.float32) / 32768.0
xt = torch.from_numpy(x)
# xt.dtype == torch.float32
```

- 16-bit PCM: discrete integer codes from the recording/ADC pipeline.
- float32 tensor: convenient numerical format for DSP and neural-network arithmetic.
- Converting int16 → float32 cannot recreate amplitude values between the original quantization levels.
- A float32 tensor can still contain badly aliased or coarsely quantized data.

More bits in the variable ≠ more bits of information in the measurement.

Live audio demo: listen in this order

The package includes pre-generated WAV files and notebook code.

- 1 **Clean 48 kHz** reference
- 2 **Naive 48→12 kHz** hear aliasing
- 3 **Filtered 48→12 kHz** compare controlled bandwidth loss
- 4 **8-bit quantized** notice noise floor
- 5 **4-bit quantized** strong granular/fuzzy error

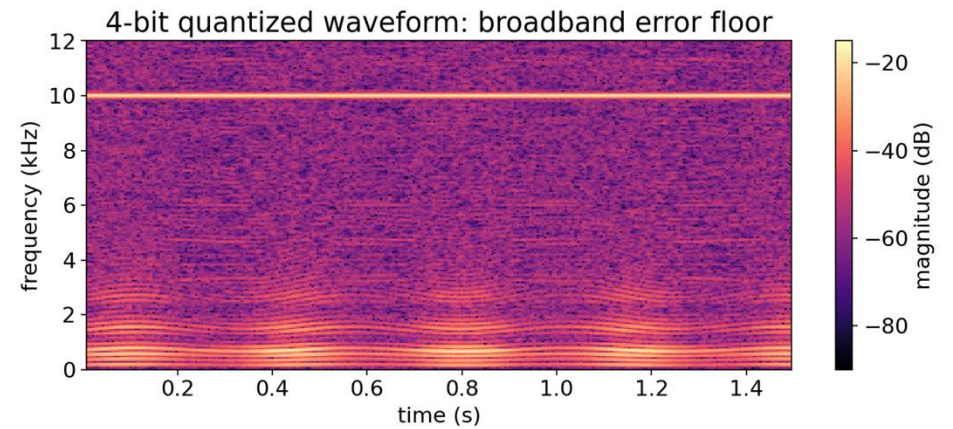
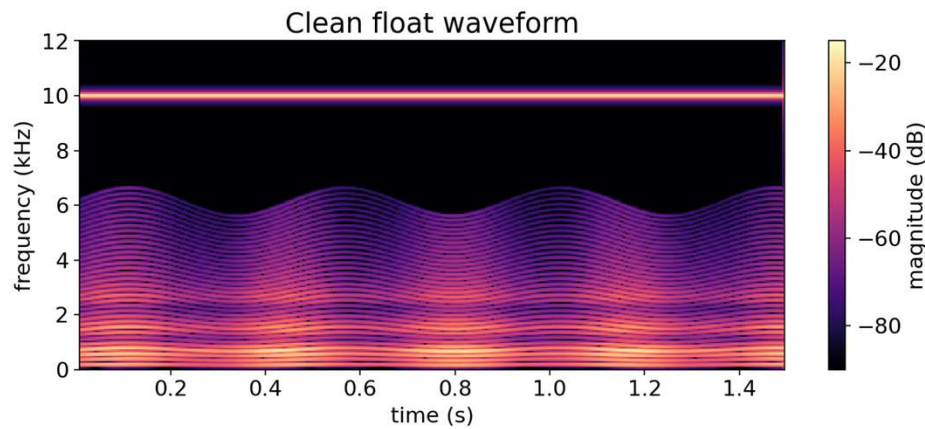
Ask students to predict what changes before playing each file.

What 4-bit quantization does to a spectrogram

The error is often broad-band, but it need not be perfectly white or independent.

Clean

4-bit



Graduate nuance: when the “white quantization noise” story breaks

Dither is one tool for trading correlated distortion for more noise-like error.

- Low-level sinusoids can produce periodic, signal-correlated error rather than white noise.
- Overload clipping is nonlinear distortion and is not captured by $\Delta^2/12$.
- Dither intentionally adds noise before quantization to decorrelate the quantization error.
- Noise shaping can push quantization noise toward frequency regions that matter less—but requires feedback/oversampling ideas beyond today.

**Modeling lesson for ML:
know which “noise
assumptions” are physics
and which are convenient
approximations.**

Exit check

Students answer individually; collect or discuss in the last two minutes.

1 Why does ideal reconstruction use sinc functions?

2 What deterministic frequency-domain effect does a ZOH introduce?

3 A full-scale 12-bit sine has approximately what ideal SQNR?

Answers: 1) sinc is the impulse response of the ideal bandlimiting reconstruction filter. 2) sinc-shaped magnitude droop (plus linear phase/delay). 3) ≈ 74 dB.

Lecture 4 summary: faithful digital audio needs both time and amplitude integrity

Sampling theory, reconstruction, and quantization are one end-to-end system.

- Ideal D/C isolates one spectrum replica and yields sinc interpolation.
- A zero-order hold is practical but imposes sinc-shaped passband droop and spectral images.
- A B-bit uniform quantizer has step $\Delta \approx 2A/2^B$; under standard assumptions $\sigma_e^2 \approx \Delta^2/12$.
- A full-scale sine gives $\text{SQNR} \approx 6.02B + 1.76$ dB.
- `int16` $\rightarrow$ `float32` is a computational conversion; it does not undo aliasing or quantization.

**Week 2 principle:
Bad acquisition creates
“features” that no
downstream ML method
can reliably repair.**